# Python Automata Library (PAL) Cheat Sheet

Full details: API Guide · API Reference

import PythonAutomataLibrary as pal · Safe mode is default. Call **pal.FastMode**() before constructing any PAL object. Put substantive model kernels under **@pal.njit(cache=True)**.

## Create state

**NewGrid**: pal.NewGrid(dims,dtype): values at lattice sites

**NewAgentGrid**: pal.NewAgentGrid(dims,numAgentProps=0,isStackable=False): individual agents

**NewPopGrid**: pal.NewPopGrid(dims,capacity=None): integer counts

**NewPDEgrid**: pal.NewPDEgrid(dims): continuous field

**NewIList**: pal.NewIList(): mutable integer list

**NewMultinomial**: pal.NewMultinomial(): binomial/multinomial draws

## Geometry / indexing

**Dimensions**: dims=(nx,), (nx,ny), or (nx,ny,nz); negative sizes wrap

**Coordinates**: grid[i] linear index; grid[x,y,z] site coordinates

**Conversions**: ToI(x,...); ItoX/Y/Z(i)

**Neighbors**: Hood(hood,x,...); pal.MooreHood, VonNeumannHood, CircleHood

**Region**: Box(...) iterates a rectangular coordinate region

## AgentGrid

**Birth**: NewAgentSQ(x,...) lattice; NewAgent(x,...) continuous

**Move**: MoveSQ(a,x,...) lattice; Move(a,x,...) continuous

**Properties**: agents[a,p] reads/writes property p

**Position**: I(a); XSQ/YSQ/ZSQ(a); X/Y/Z(a)

**Lifetime**: Alive(a); Dispose(a); GetPop()

**Queries**: AgentsAt(...); LastAgent(...); AgentsInRadius(...)

**Iteration**: All(shuffle=True) snapshots living agent handles

## Population / fields

**State**: grid[i] = v writes immediately

**Queued change**: Add(v,...) queues; Update() applies; Reset() clears

**Population**: pop.GetPop() total; pop.All() occupied indices

**PDE spacing**: field.SetTimeSpaceStep(dt,dx,dy,dz)

**Diffusion**: Diffusion, DiffusionMask, DiffusionField, DiffusionInterfaces, DiffusionADI

**Radial**: DiffusionRadialCircle / DiffusionRadialSphere

**Advection**: Advection, AdvectionField, AdvectionInterfaces

## Random / lists

**RNG**: pal.Seed(seed); pal.Random(); pal.RandInt(n)

**IList**: Append, Clear, Random, Shuffle, All (copy), Iter (no copy)

**Sampling**: m.Binomial(n,p); m.Setup(...); m.Sample(...)

## Drawing / execution

**Compile**: @pal.njit(cache=True) for model kernels

**Mode**: Safe by default; pal.FastMode() before object creation

**Pixels**: pal.StartPixWindow(...); pix[x,y] = color; win.Update()

**OpenGL**: pal.StartOpenGLWindow(...); draw.Circle/Box/Line(...)

**Output**: win.Save(path); StartGif, AddGifFrame, StopGif; Close